

each year. Many of the world's major airlines had not had a fatality in the 21st century.

The safety of commercial flying is a triumph of organization and regulation, and a tribute to the professionalism of all involved in the aviation business, from those who make the airframes, engines, and avionics, through the ground maintenance staff and flight crews to administrators and air-traffic controllers. The volume of traffic that air-traffic control has to cope with has, of course, increased dramatically in the jet age. By the late 1990s, there were some 7,000 flights a day into and out of New York. To look at it another way, controllers at Chicago's O'Hare airport were responsible for the safety of around 70 million passengers a year. But despite occasional panics about overstretched air-traffic controllers being overwhelmed by numbers, the system has continued to cope well. So has the system of periodic checks and overhauls designed to ensure that aircraft are fit to fly, with faultless engines and free of structural weaknesses. Considering what amazingly complex machines modern aircraft are—a Boeing 747 has about 4.5 million moving parts—it is astonishing how rarely they suffer serious faults. A modern jet may have ten hours ground maintenance for every hour it spends in the air.

Human error

Humans cannot be engineered to operate as faultlessly as machines. Airline pilots of course undergo rigorous training. They have mandatory medical check-ups every six months and their flying time is limited to prevent dangerous levels of stress and fatigue building up. Yet pilot error is still



WORLD'S WORST CRASH

In the aftermath of a collision between two Boeing 747s at Tenerife airport in 1977, dazed survivors try to help the injured. Images such as this, shown in the media worldwide, inevitably have more impact than the recitation of safety statistics.

held to be a contributory cause of about three quarters of air accidents. These may range from simple mistakes in reading instruments—for example flying into the ground after misreading an altimeter—to confused reactions in an emergency, as in the British Midlands Boeing 737 crash in 1989, when, with one engine failed, the crew shut down the other one, which was working.

Improvements in cockpit equipment have done a lot to aid pilots and to counter the human factor. The presentation of flight information on-screen in the new “glass cockpits,” for example, should bring an end to the simple misreading of indicators. As well as a whole gamut of navigational aids and instrument landing systems, there are now ground-proximity

CROWDED AIRPORT

Air-traffic controllers are responsible for ensuring that take-off and landing at a busy airport do not result in a collision. Airliners make a quick initial climb to carry the noise from their engines as high as possible over surrounding housing.

